Blake McAlister

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EDUCATION

Clemson University **Clemson, SC**
BS in Computer Science, Minor in Artificial Intelligence (GPA: 3.3/4.0) *Graduation Date: December 2026*

SKILLS

Programming Languages: Python, Java, JavaScript, C/C++, SQL, HTML
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Computer Vision (YOLO, OpenCV, OCR), NLP, VLMs, Model Tuning
Data Science & Analytics: NumPy, Pandas, Data Cleaning, Exploratory Data Analysis, Data Visualization, Data Modeling
Full-Stack Development: AWS, Flask, FastAPI, React, REST APIs, API Integration, Docker, Node.js, Backend Development, CI/CD
Systems: Git/GitHub, Linux/Unix, Unit & Integration Testing, High-Performance Computing (GPU, Palmetto Cluster), Cloud Computing

EXPERIENCE

Capgemini **Clemson, SC**
Industry-Sponsored AI Capstone *August 2026 - Present*

- Independently built a React/TypeScript dashboard, integrating Azure-based AI agents to deliver customer recommendations, search, and analytics across 96,000+ customer records.
- Integrated customer, purchase, and review data from Azure SQL data through REST APIs and a secure Express.js backend, contributed to recommendation agent and took ownership analytics-agent development.
- Delivered a working MVP to Capgemini stakeholders, demonstrating centralized customer analytics, search, and AI-enabled Recommendations, incorporating feedback into ongoing development to improve platform functionality and usability.

Trimble Inc. **Atlanta, GA**
Software Engineering Intern *May 2026 - August 2026*

- Built and internally demonstrated an Electron desktop prototype unifying multiple web applications while maintaining isolated user sessions.
- Shipped production frontend fixes across Trimble Pay monorepo through testing, pull requests, code reviews, and release workflows.
- Improved Gainsight PX interaction tracking by resolving multi-route conflicts with selectors, wildcard URL matching, and exclusion rules, providing more reliable product-usage data.

Watt AI Creative Inquiry Research **Clemson, SC**
Audiogram Deep Learning Researcher *August 2025 - December 2025*

- Developed a YOLO/OpenCV computer vision pipeline with image preprocessing and calibration to digitize handwritten clinical audiograms and reduce manual data entry.
- Achieved 83.1% field-level accuracy using Qwen Vision LLM for speech score extraction, evaluating OCR and Vision-LLM approaches in research involving approximately 400 audiograms.
- Built Python-based extraction workflows to convert unstructured audiogram measurements into machine-readable records, supporting more efficient clinical data analysis.

Oldcastle APG **Atlanta, GA**
Digital Solutions Intern *May 2025 - August 2025*

- Proposed an AI chatbot to automate B2B e-commerce customer inquiries, presenting its technical approach, projected ROI, and operational benefits to Oldcastle leadership, including the CEO.
- Supported BigCommerce API integrations to improve e-commerce functionality and streamline existing business workflows.
- Analyzed customer behavior through Lucky Orange to identify usability friction and recommend UX/UI improvements that enhance the online purchasing experience.

RELEVANT PROJECTS

Airline On-Time Performance Prediction **October 2026**

- Developed a machine learning pipeline to predict whether flights departing Atlanta (ATL) would arrive 15+ minutes late.
- Cleaned and analyzed 79,928 flights from U.S. DOT data, identifying delay patterns across airlines, departure times, and destinations.
- Evaluated 84 logistic regression and decision tree configurations, achieving 80% test accuracy and assessing false-negative tradeoffs.

Urba **October 2025**

- Developed a real-time environmental social media platform at NASA Space Apps Hackathon using NASA EONET and public APIs.
- Built an interactive social feed enabling users to share updates and discuss wildfires, air quality, and extreme weather.
- Implemented a city-level Wellness Score combining live environmental data and community reports to help users assess local conditions.

Vision-Based Autonomous RC Car **November 2025**

- Implemented real-time lane detection + heading correction using computer vision on a TurboPi system.
- Added car-following logic: distance estimation, stopping behavior, and restart logic for safe trailing.
- Achieved stable tracking in real hallway tests and multi-lane simulation environments.

EXTRACURRICULARS

ColorStack (Vice President) - established Clemson chapter. National Society of Black Engineers (NSBE). AI Club. Clemson WBB Practice Player. Club Basketball. Dae Han Black Belt.
Study Abroad: Mexico (May 2024): Merida, Campeche, Chichen Itza, Valladolid, Puerto Morelos.